

Can World Models Optimize Customer Journeys?

A diagnostic, simulation-based framework for sequential marketing decision-making

Working paper – draft v2

Revised: literature positioning tightened, experimental program specified, notation corrected.

Abstract. Marketing optimization has traditionally relied on predictive models that estimate customer responses to individual interventions. Uplift modeling and reinforcement learning have since pushed the field from prediction toward decision-making, and a fast-growing body of recommender-systems research has begun pairing learned simulators of user behavior with planning algorithms to reduce the cost of live experimentation. This paper asks whether that same mechanism – a learned *world model* standing in for direct interaction with customers – is a useful abstraction for marketing journey optimization specifically, where the reward is economic (revenue, margin, retention) rather than engagement, and where the open question is not whether such a system can be built, but under what conditions it can be trusted.

We propose a framework in which customer journeys are represented as partially observable, stochastic sequential decision processes. A learned world model estimates customer-state transitions conditional on marketing actions; a planning mechanism uses the learned model to compare alternative future journeys before any action is taken on a real customer. We position this framework against recent world-model-based recommender research and against LLM-based synthetic-consumer simulation, and argue that neither line of work has systematically studied the relationship between model error, planning horizon, and policy reliability – the question this paper takes as its central contribution.

To make that question tractable, we specify a staged synthetic environment with known ground-truth customer dynamics, so that the accuracy of a learned world model can be checked directly against the process that generated the data, rather than inferred indirectly from downstream policy performance. The proposed experimental program investigates transition-prediction accuracy, policy value under sequential

program investigates transition prediction accuracy, policy value under sequential versus myopic decision rules, the point at which longer planning horizons stop helping, and the sensitivity of resulting policies to deliberately injected model error.

The paper's objective is not to argue that world models are inherently superior to existing marketing models, but to establish the conditions under which this class of methods is reliable enough to inform real marketing decisions.

Keywords: world models; marketing optimization; customer journeys; reinforcement learning; model-based reinforcement learning; customer lifetime value; uplift modeling; sequential decision-making; simulation; model reliability

Contents

1. Introduction
2. From Data Mining to Sequential Marketing Optimization
3. From Prediction to Causal Marketing Decision-Making
4. From Individual Interventions to Customer Journeys
5. Customer Journeys as Dynamic Systems
6. A Reinforcement Learning Perspective
7. Model-Free versus Model-Based Decision-Making
8. World Models: Concept, Adaptation, and Latent State
9. Stochasticity
10. Planning with the World Model
11. Why Long-Term Value Matters
12. Relationship with Uplift Modeling
13. The Counterfactual Problem
14. The Problem of Model Error
15. The Marketing World Model as a Decision Simulator
16. Related Work and Positioning
17. Research Questions
18. Conceptual Propositions
19. Proposed Research Framework
20. Proposed Experimental Program
21. Expected Contributions
22. Limitations and Scope

1. Introduction

Marketing analytics has undergone a substantial evolution over the past several decades. Early data-mining applications focused on discovering associations and predicting customer behavior from historical transactions. More sophisticated approaches subsequently introduced customer-level response modeling, segmentation, personalization, recommendation, and causal uplift modeling.

A common objective has been to answer questions such as: *which customers are most likely to purchase?* or *which customers are most likely to respond positively to a particular marketing intervention?* These questions are valuable, but they describe only part of the marketing decision problem.

A customer journey is not a collection of independent decisions. Customers interact with firms through multiple touchpoints and channels, and their experiences accumulate over time. Lemon and Verhoef (2016) describe the customer journey as spanning multiple touchpoints across channels and emphasize the importance of understanding customer experience dynamically over time.

Consequently, the effect of a marketing action may extend beyond its immediate outcome. Sending a discount may increase the probability of an immediate purchase, but the same intervention may also affect future price sensitivity, brand perception, engagement, or the probability that the customer responds to subsequent communications. A model that predicts only the immediate purchase does not necessarily provide enough information to determine the optimal sequence of future interventions.

This leads to a different question:

Instead of predicting what a customer is likely to do next, can we model how the customer's state will evolve under alternative marketing actions, and use that model to optimize the journey?

This question is closely related to the concept of a *world model* in artificial intelligence: a learned representation of how an environment evolves, used to reason about possible future states before acting. Ha and Schmidhuber (2018) introduced the term in a reinforcement-learning setting in which a learned generative model captured a compressed representation of an environment and allowed policies to be trained on imagined trajectories. Model-based reinforcement-learning research developed the idea further – PlaNet (Hafner et al., 2019a) learned latent environment dynamics for online planning, demonstrating that an agent could select actions by predicting future consequences inside a learned model rather than relying exclusively on direct interaction with the environment.

This paper explores whether the same conceptual approach transfers to marketing – and, more importantly, under what conditions it is trustworthy enough to act on. As we detail in Section 16, the combination of world models and simulated user behavior is no longer purely conceptual: several 2023–2026 systems in the recommender-systems literature already build learned simulators for exactly this purpose. What remains genuinely open is not whether such a system can be assembled, but when its output should be believed.

The proposed perspective is:

Customer state → Marketing action → Future customer state → Future action → Future outcome

rather than:

Customer state → Predicted immediate response

The distinction matters because it changes the unit of optimization from the *individual intervention* to the *customer journey*. The paper therefore investigates two linked questions:

Can learned world models be used to simulate and optimize sequential marketing interventions in customer journeys – and under what conditions can the resulting policy be trusted?

The contribution is intended to be methodological and diagnostic rather than purely conceptual. We propose a controlled environment in which customer dynamics are specified explicitly, learned by a world model, and then used for planning. Because the underlying dynamics are known in the experimental environment, the approach can be evaluated not only on marketing performance but on whether it correctly learns the mechanisms governing customer evolution – and on where, precisely, it stops being safe to trust.

2. From Data Mining to Sequential Marketing Optimization

Early marketing data mining often focused on identifying relationships between observed customer behaviors. A classic example is market-basket analysis, in which transaction data are used to identify associations between products – rules of the form $\{A, B\} \rightarrow \{C\}$, where the objective is to find products or behaviors that co-occur more often than expected. This type of analysis remains useful because it transforms large volumes of transactional data into actionable patterns.

However, association is fundamentally different from intervention. Knowing that customers who purchase product A frequently purchase product B does not establish that encouraging a customer to purchase A will cause them to purchase B. This distinction becomes increasingly important as marketing systems move from analytics toward automated decision-making.

3. From Prediction to Causal Marketing Decision-Making

3.1 Response modeling

Traditional response modeling estimates the probability of a customer responding to a marketing action: $P(Y_i=1 \mid X_i)$, where X_i represents customer characteristics and historical behavior and Y_i the outcome. The resulting score can rank customers for targeting.

This approach has a fundamental limitation: a customer with a very high probability of purchasing may purchase regardless of whether the marketing intervention is

applied. Targeting that customer does not necessarily create incremental value.

3.2 Uplift modeling

Uplift modeling addresses this by estimating the incremental effect of an intervention. For a binary treatment, the individual treatment effect is

$$\tau_i = E[Y_i(1) | X_i] - E[Y_i(0) | X_i]$$

where $Y_i(1)$ is the potential outcome with the intervention and $Y_i(0)$ the potential outcome without it. Research on uplift modeling has demonstrated the value of estimating incremental treatment effects rather than simply predicting response, and multiple-treatment uplift models have been studied for selecting among promotional actions. Recent work has further shown that targeting based purely on estimated treatment effect need not correspond exactly to business-value maximization, motivating value-aware evaluation metrics.

Uplift modeling therefore represents an important transition – from *prediction* ("what will the customer do?") to *causal prediction* ("how will the customer's behavior change because of my intervention?"). But an additional problem remains.

4. From Individual Interventions to Customer Journeys

Uplift modeling generally evaluates the incremental effect of an intervention at one decision point. Customer journeys add a further dimension: *today's intervention may influence tomorrow's response*.

Consider a customer who might receive one of several interventions, $A = \{\text{email, discount, recommendation}\}$. The value of sending an email today may depend on whether the customer received an email yesterday, whether they recently purchased, whether they have become fatigued by contact, whether they are currently browsing, and whether they subsequently receive a discount.

The effect of an action is therefore not a fixed quantity – the same action, $Effect(a_t)$, can take different values depending on where in the journey it is applied and what state the customer is in when it is applied. Formally, the effect must be treated as

conditional on the evolving customer state:

$$Effect(a_t | s_t)$$

where s_t summarizes the customer's relevant state at time t . The problem consequently becomes sequential: a marketing system must determine

$$a_t^* = \arg \max_{a_t} E[\text{future value} | s_t, a_t]$$

rather than simply maximizing immediate conversion.

5. Customer Journeys as Dynamic Systems

The customer journey literature provides an important conceptual foundation here. Customer experiences develop across multiple touchpoints, channels, and interactions, and previous experiences influence subsequent ones – precisely the temporal, multi-touchpoint nature of customer journeys that Lemon and Verhoef (2016) emphasize.

A customer journey can therefore be represented as a dynamic system. At each time step t , the customer has a state s_t ; the marketing system selects an action a_t ; the customer transitions to a new state s_{t+1} and produces a reward r_t :

$$s_t \rightarrow (a_t) \rightarrow s_{t+1} \rightarrow (a_{t+1}) \rightarrow s_{t+2} \rightarrow \dots$$

This does not imply the customer is fully observable. A marketer typically observes website visits, purchases, email opens, clicks, campaign exposure, and product views, while not directly observing latent purchase intent, brand affinity, fatigue, price sensitivity, or churn probability. This makes customer-journey optimization naturally compatible with *partially observable* sequential decision-making.

6. A Reinforcement Learning Perspective

Reinforcement learning formalizes decision-making in environments where an agent

repeatedly selects actions and receives feedback. A standard formulation uses a Markov Decision Process $M = (S, A, P, R, \gamma)$, where S is the state space, A the action space, $P(s'|s,a)$ the transition function, $R(s,a,s')$ the reward function, and γ the discount factor.

RL concept	Marketing interpretation
State	Customer state
Action	Marketing intervention
Transition	Change in customer behavior / state
Reward	Revenue, margin, retention, CLV, etc.
Policy	Marketing decision strategy
Episode	Customer journey

The objective becomes maximizing cumulative expected reward, $J(\pi) = E_{\pi}[\sum_{t=0}^T \gamma^t r_t]$, which provides a natural mathematical framework for journey optimization.

Reinforcement learning applied to customer-centric marketing and lifetime-value optimization is itself not new, and – as Section 16 details more fully – is considerably more developed than a first pass through the literature suggests. Sun et al. (2023) explicitly show that individually optimal interventions can be globally suboptimal once interactions across a customer journey are taken into account. The existence of RL approaches to marketing is therefore not the research gap. The sharper question is:

Can a learned model of customer dynamics be used as an internal simulation environment for marketing planning – and how much should that simulation be trusted?

7. Model-Free versus Model-Based Decision-Making

Reinforcement learning divides broadly into model-free approaches, which learn a policy or value function directly from experience (Experience $\rightarrow$ Policy), and model-based approaches, which first learn an approximation of the environment and then

plan within it (Experience $\rightarrow$ World Model $\rightarrow$ Planning $\rightarrow$ Policy).

Model-based reinforcement learning therefore separates two problems: learning how the environment works, and determining what to do within it. Moerland et al. (2020) describe model-based RL as integrating learned environment dynamics with planning, while highlighting stochasticity, uncertainty, partial observability, and long-horizon prediction as central challenges.

For marketing, this decomposition is attractive because direct experimentation with customers is often expensive or undesirable. If a sufficiently accurate model of customer dynamics can be learned, alternative strategies could in principle be evaluated *inside* the model before deployment – which is exactly the appeal that has drawn recommender-systems research toward the same idea (Section 16).

8. World Models: Concept, Adaptation, and Latent State

8.1 What is a world model?

The term does not have a single fixed definition. Broadly, a world model is a learned representation of an environment capturing enough of its dynamics to support prediction, simulation, or planning. Ha and Schmidhuber's (2018) formulation used a generative model to learn compressed spatial and temporal representations of an environment, letting agents operate within imagined versions of it. PlaNet (Hafner et al., 2019a) showed latent dynamics could be learned and used for planning; Dreamer (Hafner et al., 2019b) then learned behavior through latent imagination generated by the world model itself; DreamerV3 (Hafner et al., 2023) extended the approach across a broad set of tasks with a single general architecture. The concept that matters for this paper is not any specific architecture, but the separation between *the environment* and *a learned internal model of the environment*.

8.2 Adapting the concept to marketing

Let the true customer dynamics be $s_{t+1} \sim P(s_{t+1} | s_t, a_t)$. The marketer does not know P ; instead, historical customer journeys are used to learn an approximation $\hat{P}_\theta(s_{t+1} | s_t, a_t)$ – the *customer world model*. This model can generate hypothetical trajectories, $s_t \rightarrow a_t$

$\rightarrow \tilde{s}_{t+1} \rightarrow a_{t+1} \rightarrow \tilde{s}_{t+2} \rightarrow \dots$, letting a decision-maker compare alternative sequences without applying every sequence to real customers – for instance, comparing an email-then-recommendation journey against a discount-then-email journey, or against taking no action at all, purely inside the simulator.

The contrast with a conventional predictive model is worth making explicit. A predictive model estimates an immediate outcome, $P(\text{Purchase}_{t+1} \mid X_t, A_t)$ – "a discount increases purchase probability by 8%." A world model estimates the broader transition, $P(S_{t+1} \mid S_t, A_t)$ – "if I discount this customer now, how does their purchase probability change today, how does their state change afterward, and how does that reshape their response to whatever I do next?" The second question is inherently sequential; the first is not.

8.3 Latent customer state

A simple state representation might be $s_t = [E_t, P_t, F_t, L_t, C_t]$ – engagement, purchase propensity, marketing fatigue, loyalty, and churn propensity. These variables need not be directly observable. The world model may instead learn a latent representation $z_t = f_\theta(h_t)$ from the history of observations and actions h_t , with transitions $z_{t+1} \sim P_\theta(z_{t+1} \mid z_t, a_t)$ – the same approach used by latent world models such as PlaNet, where the model need not reproduce every observable detail, only what is useful for predicting outcomes and planning. For marketing, this is potentially advantageous precisely because many important customer characteristics – intent, fatigue, affinity – are inherently latent.

9. Stochasticity

Customer behavior is not deterministic: two customers with apparently identical observable characteristics may respond differently to the same intervention. A deterministic transition, $s_{t+1} = f(s_t, a_t)$, is therefore generally insufficient; a more realistic formulation is $s_{t+1} = f(s_t, a_t, \varepsilon_t)$, where ε_t captures stochastic variation, and the world model should ideally represent the full distribution $P(s_{t+1} \mid s_t, a_t)$ rather than a single point prediction.

This matters for decision-making: a model that predicts only "customer will purchase"

invites an overconfident planner, whereas a probabilistic model can represent "35% probability of purchase, 50% probability of remaining engaged, 15% probability of disengaging." Uncertainty then becomes part of the decision problem itself, rather than an error term to be minimized away.

10. Planning with the World Model

Given a current state s_t , a planner evaluates candidate action sequences $(a_t, a_{t+1}, \dots, a_{t+H})$ over a planning horizon H . The world model generates simulated future states $\hat{s}_{t+1}, \dots, \hat{s}_{t+H}$ and rewards $\hat{r}_{t+1}, \dots, \hat{r}_{t+H}$; the planner evaluates

$$\hat{J} = \sum_{k=0}^H \gamma^k \hat{r}_{t+k}$$

and selects the sequence with the highest expected value. In practice, the system would normally execute only the next action and re-estimate the customer state once new information arrives – a *receding-horizon* loop: observe $\rightarrow$ simulate $\rightarrow$ choose $\rightarrow$ act $\rightarrow$ observe $\rightarrow$ simulate again. This is closely related to model-predictive control and decision-time planning in model-based reinforcement learning.

11. Why Long-Term Value Matters

Marketing optimization is often evaluated through immediate metrics – click-through rate, conversion, revenue, campaign response. But customer-level optimization can also involve longer-term objectives such as customer lifetime value. Kumar and George (2007) emphasize the importance of customer-level resource allocation and balancing acquisition against retention when maximizing customer equity.

This creates a natural distinction between a short-term objective, $\max P(\text{Purchase}_{t+1})$, and a long-term one, $\max E[\sum_{t=0}^T \gamma^t \text{Value}_t]$. The optimal intervention under these two objectives may differ: an aggressive discount can maximize immediate conversion while eroding future margin or deepening a customer's dependence on promotions. A world model creates a mechanism through which such delayed effects can, in principle, be represented and optimized rather than discovered too late.

12. Relationship with Uplift Modeling

World models should not be read as a replacement for uplift modeling; the two address different levels of the problem. Uplift modeling asks what the incremental effect of this intervention on this customer is; a world model asks how the customer's state evolves after this intervention and the interventions that follow it. Uplift modeling is therefore a useful building block for a sequential decision system – an immediate causal effect – while a world model targets dynamic causal and behavioral simulation.

This distinction also surfaces a methodological challenge: a world model trained purely on observational data does not automatically learn causal effects. If historical campaigns were targeted non-randomly, the model may learn correlations rather than intervention effects. A marketing world model intended for decision-making should therefore ideally be trained on experimental, quasi-experimental, or otherwise causally informative data – a concern we return to concretely in Section 20.2, where it becomes a manipulated variable rather than a caveat.

13. The Counterfactual Problem

Marketing optimization is fundamentally counterfactual. For a customer who received a discount and purchased, we observe $Y(1) = 1$ but never $Y(0)$ – the outcome that would have occurred without the discount. This is the standard challenge addressed by causal inference and uplift modeling.

The sequential version is considerably harder. The question is no longer "what would have happened without this one intervention?" but "what would have happened if we had sent an email instead of a discount, followed by a recommendation two days later, followed by no communication?" – a rapidly expanding counterfactual space. A world model offers one mechanism for navigating that space by learning the underlying dynamics and generating simulated trajectories, but simulation does not eliminate the causal-identification problem; it simply relocates it, from estimating one treatment effect to learning a model capable of generating *credible* counterfactual trajectories.

This distinction is essential to avoid overstating what a world model can achieve.

14. The Problem of Model Error

The central risk of model-based planning is that an optimizer can exploit inaccuracies in its own model. If the true environment is P and the learned world model is $\hat{P}$, the planner optimizes $\pi^* = \arg \max_{\pi} J(\pi | \hat{P})$ rather than $\arg \max_{\pi} J(\pi | P)$. If $\hat{P}$ is inaccurate, the resulting policy may be optimal *inside the model* but poor in the real environment – and this risk grows with the planning horizon, since small transition errors compound across simulated steps. Moerland et al. (2020) identify model uncertainty and long-horizon prediction as central open challenges in model-based RL generally.

For marketing, this has an intuitive reading: the optimizer may discover a customer journey that looks highly profitable according to the simulated customer, but would not actually produce that behavior in reality. The relationship between world-model accuracy and policy quality is therefore one of this paper's central questions, and – unlike in most of the prior work surveyed in Section 16 – we treat it as the thing to be measured, not assumed away.

15. The Marketing World Model as a Decision Simulator

The proposed framework can be summarized as five layers, in a loop:

Layer 1 – Customer observations (*transactions, interactions, campaign exposure, browsing, engagement, churn, purchases*)



Layer 2 – Customer state representation (*observed and latent characteristics*)



Layer 3 – World model, learns $P(s_{t+1} | s_t, a_t)$



Layer 4 – Planning (*simulates alternative future journeys*)



Layer 5 – Marketing policy (*selects the action expected to maximize the objective*)

The complete loop is: customer data → state estimation → world model → future simulation → policy optimization → marketing action → new customer observation → state update → repeat.

16. Related Work and Positioning

The world-model framing proposed here needs to be positioned against a fast-moving adjacent literature in recommender systems and applied reinforcement learning, most of it published within the last two to three years – later than the world-models and marketing-RL literature the earlier sections draw on, and easy to miss on a first pass.

Several recent systems already combine learned world models with simulated user feedback for policy learning. Li et al. (2026) propose a framework in which a "User Engagement World Model" acts as a frozen simulator supplying reward signal before a policy is exposed to real users, motivated explicitly by the cost, latency, and risk of live feedback collection and by the lack of counterfactual comparisons under a shared user state. Bougie et al. (2026) use world models to align LLM-driven simulated users with real behavior for recommender-system evaluation. Chen et al.'s (2023) Sim2Rec applies simulator-based reinforcement learning specifically to long-term user engagement in sequential recommenders – a paper contemporaneous with DreamerV3 that addresses the same exploration cost motivating the present framework. On the environment side, Xia, Arian, Narayanamoorthy, and Mabry (2023) introduced RetailSynth, a multi-stage generative model of retail shopping behavior calibrated against public grocery transaction data, later used to benchmark contextual-bandit and deep-RL coupon-targeting agents across a full retail customer journey (Xia, Narayanamoorthy, Zhou, & Mabry, 2024). Separately, Singh et al. (2025) extend model-based RL agents with a *reverse* world model to generate counterfactual explanations of agent behavior, which is further evidence that "world model" is now an actively used term in applied RL well beyond games and robotics.

These contributions establish that the core mechanism proposed in this paper – a learned simulator standing in for costly real interaction – is no longer a purely

learned simulator standing in for costly real interaction — is no longer a purely conceptual proposal. This paper's positioning is therefore narrower than "no one has combined world models with sequential decision-making over user behavior": the combination exists, but (a) none of it is framed around marketing economics — all optimize an engagement- or click-style reward rather than margin, discount cost, or retention value — and (b) none of it systematically studies the conditions under which the learned model can be trusted. Model error, planning horizon, and the causal identifiability of the training data are treated as implementation details in this literature, not as the object of study. The present paper takes the reliability question itself as its primary contribution, which is also why its experimental program (Section 20) is built around a synthetic environment with known ground truth rather than a benchmark on realistic-looking data: the point is to be able to check the world model's answers against the process that actually generated them.

A second, related literature uses large language models rather than parametric simulators to generate synthetic respondents. Park et al. (2024) digitally reconstruct the attitudes and behaviors of over a thousand real people to study how a diverse population might respond to new scenarios. Chu et al. (2025) build a multi-agent LLM simulation of consumer decision-making to pre-test a price-discount marketing scenario, reporting emergent social patterns that rule-based agent-based models do not capture. Independent market-research evaluations reported in industry trade press have found LLM-generated synthetic respondents approaching human test-retest reliability on some product categories, though results vary by category and elicitation method. This line of work targets behavioral realism and external validity rather than the causal transparency that a known-ground-truth simulator provides, and we treat it in this paper as a natural extension rather than a substitute — see Section 22.

17. Research Questions

RQ1 – Dynamics. Can a learned world model accurately represent the dynamics of customer-state transitions following marketing interventions?

RQ2 – Optimization. Can a planner using the learned world model identify marketing strategies that improve cumulative customer value compared with non-sequential decision approaches?

RQ3 – Long-term effects. Does explicit modeling of customer-state transitions produce different decisions from approaches optimized primarily for immediate response?

RQ4 – Planning horizon. How does the quality of marketing decisions change as the planning horizon increases?

RQ5 – Model uncertainty. How does world-model uncertainty and prediction error affect the reliability of the resulting marketing policy?

18. Conceptual Propositions

P1. If customer behavior contains meaningful temporal dependencies, a model that explicitly represents state transitions should provide more information for sequential optimization than a model focused exclusively on immediate response. *Operationalized in Section 20.5 as the gap between transition-prediction error and single-step response-model error.*

P2. The value of world-model-based planning should increase when marketing actions have delayed or persistent effects. *Operationalized as the policy-value gap between sequential and myopic policies, varied against the strength of the engineered delayed effect in the environment.*

P3. The advantage of long-horizon planning should decrease when world-model prediction error becomes sufficiently large. *Operationalized as policy value against planning horizon H , at fixed model accuracy.*

P4. A planner optimized against an inaccurate world model may produce policies that perform well in simulation but poorly in the underlying environment. *Operationalized as simulated policy value versus true-environment policy value, under deliberately injected model bias.*

P5. World-model-based optimization should be most useful when the business objective is inherently cumulative, such as customer lifetime value, rather than purely immediate, such as single-period conversion. *Operationalized as the advantage of model-based planning as the discount factor γ approaches 1.*

These propositions will be evaluated experimentally rather than treated as established facts.

19. Proposed Research Framework

Customer history → State representation → World model → Future simulation →
Planning → Marketing action

with a feedback loop: marketing action → customer response → new state → world model. The framework represents marketing as a closed-loop adaptive system rather than a sequence of isolated campaigns.

20. Proposed Experimental Program

Rather than leaving the experimental design open, we specify a concrete, staged program built around a synthetic environment with *known* ground-truth dynamics – the principle proposed in Section 19, developed to the point of being directly implementable.

20.1 Environment: two levels of difficulty

Level 0 – fully observable (primary experiment). Customer state $s_t = (E_t, F_t, P_t)$ – engagement, marketing fatigue, and purchase propensity, each bounded in $[0,1]$. Action set $A = \{\text{none, email, discount, recommendation}\}$. Transitions follow a simple logistic update with additive noise, deliberately calibrated so that one action (discount) raises immediate purchase probability while also raising future fatigue – an engineered delayed cost, without which there is nothing for sequential planning to discover that a single-step model would miss. Reward combines purchase revenue, discount cost, and a small penalty for unsubscribing once fatigue crosses a threshold.

Level 1 – partially observable (robustness check). An unobserved discrete segment variable modulates the Level-0 transition coefficients; the marketer observes only noisy proxies of engagement and propensity. This is where a recurrent or attention-based encoder becomes necessary rather than optional, and where the latent-state claims of Section 8.2 become testable rather than assumed.

Level 0 is run to completion first; Level 1 is attempted only once the Level-0 pipeline – data generation, world-model fitting, planning, evaluation – is validated end to end.

20.2 Data generation

Trajectories are logged under two policies: (a) a fully randomized policy, giving unconfounded but exploratory coverage, and (b) a confounded, rule-based "business as usual" policy (e.g., discounting only customers already flagged as low-propensity), mimicking how real marketing logs are actually collected. World models are trained separately on each, turning the causal-identification concern of Section 13 into a measured effect rather than a caveat.

20.3 World model and planning

Model complexity is added only as the environment requires it: a linear or shallow MLP transition model for Level 0, moving to a recurrent latent-state model only once Level 1 introduces history-dependence. With a small, discrete action space, planning uses short-horizon Monte Carlo rollout search inside the learned model rather than a learned neural planner, keeping the relationship between planning inputs and outputs interpretable – deliberately deferring the architecture choice the earlier draft of this framework left open, until the environment actually forces the question.

20.4 Baselines

- Random policy.
- Single-step / myopic uplift policy – an upper bound on what response modeling alone can achieve.
- World-model planner trained on randomized logs.
- World-model planner trained on confounded logs.
- Oracle planner, which plans against the *true* transition function rather than a learned one – isolating planning-algorithm error from world-model error.
- Model-free RL trained with unlimited access to the true simulator – an upper bound that is not achievable with real customers, reported as a ceiling rather than a competitor.

20.5 Metrics and proposition mapping

Proposition	Manipulated variable	Measured outcome
P1	Model class (single-step vs. transition)	Transition-prediction error vs. myopic response-model error
P2	Strength of the fatigue $\rightarrow$ propensity link	Policy-value gap: sequential vs. myopic
P3	Planning horizon H	Policy value vs. H , at fixed model accuracy
P4	Injected model bias	Simulated policy value vs. true-environment policy value
P5	Discount factor γ	Advantage of model-based planning as $\gamma \rightarrow 1$

20.6 What Level 0 alone can already answer

Even before Level 1 is built, the Level-0 program directly answers RQ1 (dynamics), RQ2 (optimization), and RQ3 (long-term effects), and – by sweeping planning horizon and injected model bias – RQ4 and RQ5 as well. Level 1 exists to test whether the same conclusions survive once the customer state is no longer fully known, which is a distinct and harder question that should not be conflated with the core result.

21. Expected Contributions

Conceptual. The paper frames marketing journey optimization as a world-model-based sequential decision problem, explicitly positioned against the recommender-systems literature that has begun exploring the same mechanism under an engagement rather than a marketing-economics objective.

Methodological. It proposes a staged, known-ground-truth environment and a causal-identification experiment (randomized vs. confounded logging) for evaluating whether learned customer dynamics can support counterfactual journey simulation and marketing planning, with each conceptual proposition mapped to a specific, runnable comparison.

Practical. It aims to identify the conditions – expressed in terms of planning horizon and model error, not a general endorsement – under which simulated customer journeys provide useful decision support, and the conditions under which they do not.

The paper therefore avoids the simplistic conclusion that "world models are better for marketing." The intended contribution is to determine *when, why, and under what conditions* this class of methods can provide useful marketing optimization.

22. Limitations and Scope

- A simulated customer environment cannot reproduce the full complexity of real consumer behavior.
- The validity of a learned world model depends on the quality and causal informativeness of the data it is learned from.
- Customer behavior is affected by variables that are difficult to observe, including competitors, macroeconomic conditions, social influences, and shifting preferences.
- The reward function necessarily reflects a particular business objective; optimizing revenue, margin, retention, or lifetime value may lead to different policies.
- A model can generate internally coherent trajectories without those trajectories being realistic – predictive plausibility must be distinguished from causal validity.
- Successful simulation does not imply a policy should be deployed directly against real customers; deployment would require additional validation, experimentation, governance, and monitoring.
- The LLM-based synthetic-respondent literature discussed in Section 16 is a natural next step for testing whether these conclusions generalize to richer, less-controlled behavior generators, but it is explicitly out of scope here: the core requirement of this study – a known ground truth to check the world model against – does not hold for LLM-generated personas.

23. Discussion

The central conceptual shift proposed by this research is from *marketing prediction* to *marketing simulation*. Traditional analytics asks who will buy; uplift modeling asks who will change behavior because of an intervention; sequential decision-making asks what to do next. A world-model approach adds a further question: *what might happen to the customer after I do it?*

This allows marketing optimization to be expressed as a dynamic system, which matters increasingly as marketing organizations move toward automated journey orchestration – an automated system capable of selecting actions at scale needs more than accurate predictions; it needs a mechanism for reasoning about the consequences of actions. A world model is one such mechanism, but it carries a fundamental tension: the more aggressively the system plans into the future, the more it depends on the accuracy of its internal representation of the customer. A longer horizon brings more potential value and more model uncertainty at once. This is why the research treats *uncertainty*, rather than accuracy alone, as the central consideration – and why, as Section 16 argues, the field's open problem is no longer whether a world-model-based marketing system can be built, but whether its answers can be trusted, and under what conditions.

This distinction between predictive fidelity and decision utility is fundamental: a world model that predicts individual transitions only approximately may still produce useful policies if it captures the dimensions of customer state that matter for decisions, while a model that fits transitions well can still produce harmful policies if small errors compound over a long planning horizon.

24. Conclusion

Marketing optimization has progressively moved from descriptive analytics toward prediction, causal targeting, and sequential decision-making. The next potential step is to model not only customer responses but the evolution of the customer itself. This paper proposes learned world models as a framework for doing so, representing the relationship

Customer state + Marketing action → Future customer state

Once learned, this model can simulate alternative customer journeys and evaluate potential sequences of marketing interventions, changing the optimization problem from *which customer should receive this campaign?* to *which sequence of interventions is most likely to create the desired long-term customer outcome?*

This connects customer journey modeling, causal marketing, uplift modeling, reinforcement learning, and world-model-based planning – several of which, as Section 16 shows, are already being connected by adjacent recommender-systems research. The critical empirical question is therefore not whether such a framework can be constructed, but whether the learned model is sufficiently accurate and robust to support useful decisions, and precisely where that stops being true. The proposed experimental program focuses on the chain

World-model accuracy → Simulation quality → Planning quality → Marketing performance

Understanding this chain, under conditions where the ground truth is known and can be checked, is what will determine whether world models represent a meaningful new approach to marketing optimization, or simply another form of predictive modeling wearing a more ambitious name.

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Additional literature on causal inference, sequential recommendation, and CLV estimation should be incorporated during the systematic literature-review pass.